

Probability of Default (PD) Modeling Report

Credit Risk Binary Classification

This report summarizes an end-to-end credit-risk modeling workflow using the dataset included in the repository under `data/raw/credit_risk.csv`.

Problem Statement

Build a binary classification model that estimates the probability a borrower will default on a loan and supports risk ranking, manual review, and portfolio monitoring.

Approach

The final workflow predicts loan outcome Status while using prior Default history as a predictor. Preprocessing, imputation, encoding, feature engineering, model fitting, and threshold selection are handled in leakage-safe sklearn pipelines.

Model Comparison

Model	ROC-AUC	PR-AUC	F1-score
Logistic Regression	0.8649	0.7107	65.26%
Random Forest	0.9195	0.8605	78.17%
Hist Gradient Boosting	0.9342	0.8824	79.89%
XGBoost Baseline	0.9390	0.8856	79.62%
Final Tuned XGBoost	0.9477	0.8994	81.51%

Final Model Performance

Metric	Held-out Test Result
ROC-AUC	0.9477
PR-AUC	0.8994
Accuracy	92.70%
Precision	91.06%
Recall	73.77%
F1-score	81.51%
KS Statistic	0.7572
Brier Score	0.0656

Interpretation and Calibration

Top predictive factors include home ownership, loan-to-income ratio, interest rate, loan intent, prior default history, income, and percent income used by the loan. Sigmoid calibration improved probability quality, reducing Brier score from about 0.0704 to about 0.0597.

Business Implications

The model can support manual review prioritization, risk-based pricing, policy design, and portfolio monitoring. It should be used with human oversight, fairness monitoring, and periodic retraining.